

**▪ Problem 4 (15/01/2021)**

A steel workpiece ( $k_{cs} = 2800$  MPa, Kronenberg coefficient  $x = 0,2$ ) must be machined on a planar surface with a length of 600 mm and a width of 20 mm. The operation involves a roughing step with a peripheral milling operation characterized by a machining allowance equal to 2 mm. The selected tool has a diameter of 20 mm, a working height of 30 mm and 4 straight cutting edges. The process parameters used are so that the feed velocity  $v_f$  is equal to 384 mm/min, and the number of revolutions per minutes  $n$  is equal to 1200 rpm.

Determine the chip thickness, the cutting force, the cutting torque, and the cutting power, verifying that the operation can be performed with a milling machine with a power of 7,5 kW and characterized by an efficiency of 0,9.



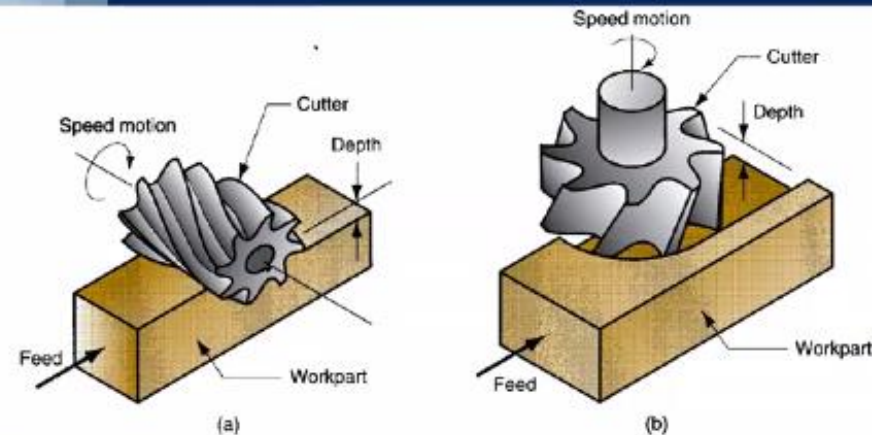
### ■ Problem 4 (15/01/2021)

A steel workpiece ( $k_{cs} = 2800$  MPa, Kronenberg coefficient  $x = 0,2$ ) must be machined on a planar surface with a length of 600 mm and a width of 20 mm. The operation involves a roughing step with a **peripheral milling operation** characterized by a machining allowance equal to 2 mm. The selected tool has a diameter of 20 mm, a working height of 30 mm and 4 straight cutting edges. The process parameters used are so that the feed velocity  $v_f$  is equal to 384 mm/min, and the number of revolutions per minutes  $n$  is equal to 1200 rpm.



#### TWO FORMS OF MILLING

5



Peripheral milling (a)

- Cutter axis parallel to surface being machined
- Cutting edges on outside periphery of cutter

Face milling (b)

- Cutter axis perpendicular to surface being milled
- Cutting edges on both the end and outside periphery of the cutter



### ▪ **Solution**

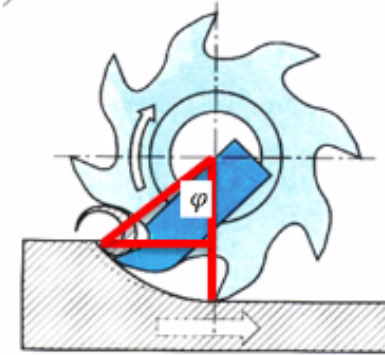
In order to answer, it is necessary to understand what the actual situation in this **slab milling operation** is.

First of all, we need to compute the engagement angle and then the number of simultaneously working teeth:

$$\varphi = \arccos \frac{\frac{D}{2} - a_e}{\frac{D}{2}} = \arccos \frac{\frac{20}{2} - 2}{\frac{20}{2}} = \arccos 0,8 = 36,9^\circ = 0,644 \text{ rad}$$

$$\varphi_0 = \frac{2\pi}{Z} = \frac{2\pi}{4} = 1,571 \text{ rad} = 90^\circ$$

$$z = \frac{\varphi}{\varphi_0} = \frac{0,644}{1,571} = 0,41$$



Therefore, we have to consider **the worst-case scenario** when a **single cutting edge is cutting the maximum chip thickness**, that we have when the angular position of the cutting edge is equal to the engagement angle.

In order to do the computation, we need to determine the feed per tooth from the data.

$$n = 1.200 \text{ RPM}$$

$$v_f = 384 \frac{\text{mm}}{\text{min}}$$

$$v_f = Z \cdot f_z \cdot n \rightarrow f_z = \frac{v_f}{Z \cdot n} = \frac{384}{4 \cdot 1.200} = 0,08 \frac{\text{mm}}{\text{rev}}$$



▪ **Solution**

Therefore:

$$h_{D,max} = h_D(\theta = \varphi) = f_Z \sin \varphi = 0,08 \cdot \sin 36,9^\circ = 0,048 \text{ mm}$$

$$A_{D,max} = A_D(\theta = \varphi) = h_{D,max} \cdot a_p = 0,048 \cdot 20 = 0,96 \text{ mm}^2$$

$$k_{c,max} = k_{cs} h_{D,max}^{-x} = 2.800 \cdot 0,048^{-0,2} = 5.139,4 \text{ Mpa}$$

$$F_{c,max} = k_{c,max} A_{D,max} = 5.139,4 \cdot 0,96 = 4.934 \text{ N}$$

$$T_c = F_{c,max} \frac{D/2}{1000} = 4.934 \frac{20/2}{1000} = 49,34 \text{ Nm}$$

$$\omega = \frac{2\pi n}{60} = \frac{2\pi \cdot 1.200}{60} = 125,67 \frac{\text{rad}}{\text{s}}$$

$$P_c = T_c \omega = 49,34 \cdot 125,67 = 6.200 \text{ W} = 6,2 \text{ kW}$$

$$P_{c,limit} = E \cdot P_e = 0,9 \cdot 7,5 = 6,75 \text{ kW}$$

Therefore, being  $P_c = 6,2 \text{ kW} \leq P_{c,limit} = 6,75 \text{ kW}$ , the slab milling operation is feasible.



**▪ Problem 5 (10/09/2021)**

A planar surface of a steel workpiece ( $k_{cs} = 2800$  MPa, Kronenberg coefficient  $x = 0,2$ ) must be machined. The planar surface has a rectangular shape with a length of 1000 mm and a width of 50 mm. The peripheral milling operation involves a roughing step characterized by a machining allowance equal to 5 mm. The selected tool has a diameter of 63 mm, a working height of 110 mm, and 4 cutting edges. The process parameters used are so that the feed velocity is equal to 480 mm/min, and the number of revolutions per minutes is equal to 600.

Determine the chip thickness, the cutting force, the cutting torque, the cutting power, and the cutting time, verifying that the operation can be performed with a milling machine with a power of 60 kW and characterized by an efficiency of 0,9.



### ▪ **Solution**

To answer, it is necessary to understand what the actual situation in this slab milling operation is.

First, we need to compute the engagement angle and then the number of simultaneously working teeth:

$$\varphi = \arccos \frac{\frac{D}{2} - a_e}{\frac{D}{2}} = \arccos \frac{\frac{63}{2} - 5}{\frac{63}{2}} = \arccos 0,84 = 32,86^\circ = 0,574 \text{ rad}$$

$$\varphi_0 = \frac{2\pi}{Z} = \frac{2\pi}{4} = 1,571 \text{ rad} = 90^\circ$$

$$z = \frac{\varphi}{\varphi_0} = \frac{0,574}{1,571} = 0,37$$

Therefore, we must consider the worst-case scenario when a single cutting edge is cutting the maximum chip thickness, that we have when the angular position of the cutting edge is equal to the engagement angle.

To do the computation, we need to determine the feed per tooth from the data.

$$n = 600 \text{ RPM}$$

$$v_f = 480 \frac{\text{mm}}{\text{min}}$$

$$v_f = Z \cdot f_z \cdot n \rightarrow f_z = \frac{v_f}{Z \cdot n} = \frac{480}{4 \cdot 600} = 0,2 \frac{\text{mm}}{\text{rev}}$$



▪ **Solution**

Therefore:

$$h_{D,max} = h_D(\theta = \varphi) = f_Z \sin \varphi = 0,2 \cdot \sin 32,86^\circ = 0,109 \text{ mm}$$

$$A_{D,max} = A_D(\theta = \varphi) = h_{D,max} \cdot a_p = 0,109 \cdot 50 = 5,45 \text{ mm}^2$$

$$k_{c,max} = k_{cs} h_{D,max}^{-x} = 2.800 \cdot 0,109^{-0,2} = 4.362 \text{ Mpa}$$

$$F_{c,max} = k_{c,max} A_{D,max} = 4362 \cdot 5,45 = 23.773 \text{ N}$$

$$T_c = F_{c,max} \frac{D/2}{1000} = 23.773 \frac{63/2}{1000} = 748,8 \text{ Nm}$$

$$\omega = \frac{2\pi n}{60} = \frac{2\pi \cdot 600}{60} = 62,83 \frac{\text{rad}}{\text{s}}$$

$$P_c = T_c \omega = 748,8 \cdot 62,83 = 47.047 \text{ W} = 47 \text{ kW}$$

$$P_{c,limit} = E \cdot P_e = 0,9 \cdot 60 = 54 \text{ kW}$$

Therefore, being  $P_c = 47 \text{ kW} \leq P_{c,limit} = 54 \text{ kW}$ , the slab milling operation is feasible.

The cutting time in the slab milling operation is equal to:

$$t_c = \frac{L + \frac{D}{2} \sin \varphi}{v_f} = \frac{1000 + \frac{63}{2} \sin 32,86^\circ}{480} = 2,12 \text{ min.}$$

